Build Overview:

1 **PSU Manifold & Lines Install**
2 **CDU Install**

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

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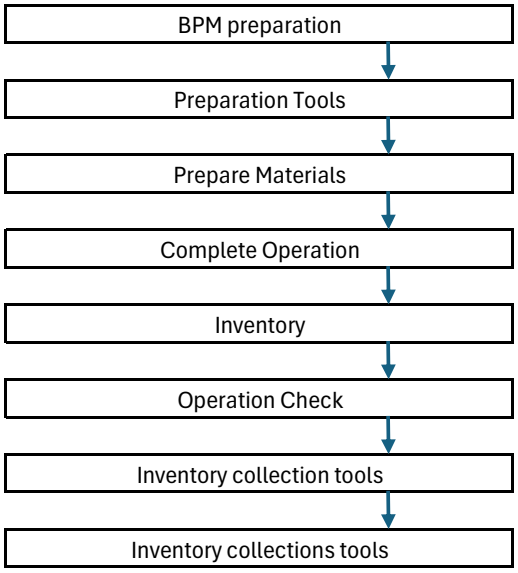
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM NO.	PART NUMBER	PART NAME
1	633-694-01	PSU MANIFOLD
2	614-766-90	LOCK BLOCKS
3	470-502-07	M8 SPRING WASHER
4	470-501-06	M8 FlAT WASHER
5	470-933-01	M8 SCREW
6	472-426-12	M5 SEMS SCREW
7	639-582-01	STEEL FLEXIBLE TUBING (UPPER JOINT)
8	639-583-01	STEEL FLEXIBLE TUBING (LOWER JOINT)
9	634-175-01	DOUBLE TUBING CLAMP
10	602-621-01	SCREW, CAP BUT HD M8X1.25PX45MML SS
11	631-823-00	CDU "Z" bracket
12	472-426-20	M5 Long Screws
13	629-538-00	ADB Shelf Bracket

METHOD:

Flow chart of operation steps



MACHINE:

1. Torque Wrench 3.3Nm, 13.5Nm, 1.6Nm, 6.8Nm.
2. Bit PH2, Hex 5mm, 1 Inch Adapter
3. 1"Open End Wrench
4. Long Foam Jig, CDU Lifter,

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	633-694-01	PSU Manifold	1

MACHINE: Torque 13.5Nm & 3.3Nm, Bit Hex 5mm, PH2

METHOD: **PSU Manifold Preparation**

1. Take 1 Manifold (633-694-01) as shown in the picture

A. Make sure all QD have torque marking and all QD are tighten.

MAN: CERTIFIED OPERATOR, TECHNICIAN

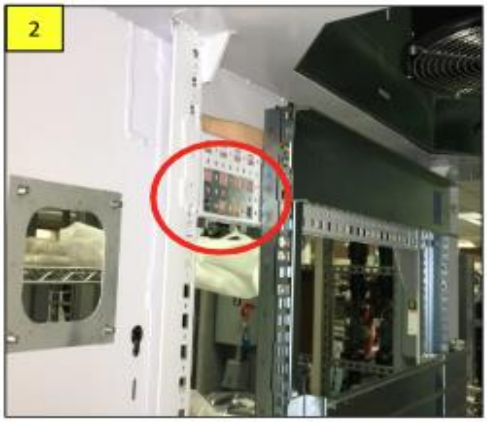
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	617-766-90	Lock Block	1

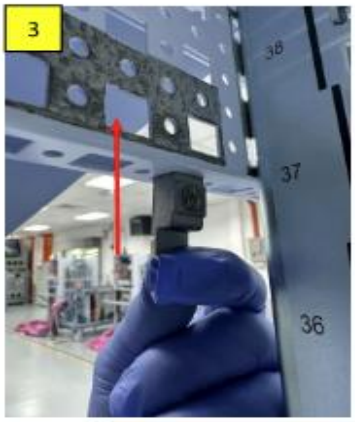
MACHINE: Torque 13.5Nm & 3.3Nm, Bit Hex 5mm, PH2

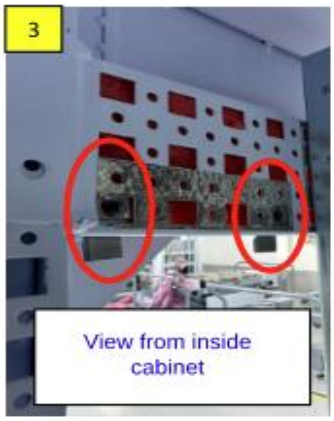
METHOD: **Lock Block Installation**

1. Take 1 Manifold (633-694-01) as shown in the picture



2. On **Operation Side** of cabinet, install 2pcs Lock Block(614-766-90) at the upper left inside the cabinet.





3. Insert 2pcs Lock Block (614-766-90) as shown in figure and hold it with Long Foam Jig.

MAN: CERTIFIED OPERATOR, TECHNICIAN

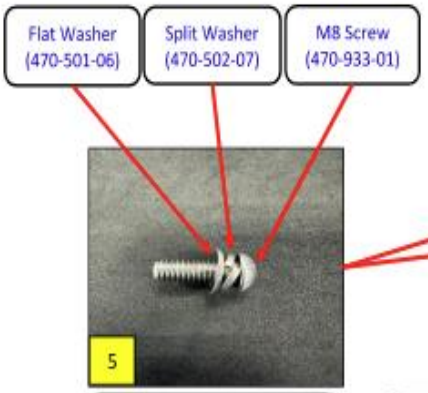
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	470-502-07	M8 Spring Washer	2
2	470-501-06	M8 Flat Washer	2
3	470-993-01	M8 Screw	2
4	472-426-12	M5 Sems Screw	1

MACHINE: Torque 13.5Nm, 3.3Nm, Bit Hex 5mm, PH2, Long Foam Jig

METHOD: **PSU Manifold Installation**

4. Install the PSU Manifold (633-694-01) into cabinet
Note : The PSU Manifold inserted from outside of the cabinet.



5. Diagram above shown orientation of screw. 2 set screw needed.

6. Secure the PSU Manifold (633-694-01) with 2 set screw that have made from previous process. Tighten the screw until the PSU Manifold can hold to cabinet.

7. Insert 1pcs M5 Screw (472-426-12) at location **28M**

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: Torque 13.5Nm & 3.3Nm, Bit Hex 5mm, PH2

METHOD: **PSU Manifold Installation**

8. Tighten 2 pcs screw with 13.5Nm torque and Hex 5mm Bit

9. Tighten 1pcs screw with 3.3Nm torqueand PH2 Bit

IMPORTANT NOTE : After all PSU Manifold installation proces have done, Take Out the Long Foam Jig use before

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 1 Inch Open End Wrench

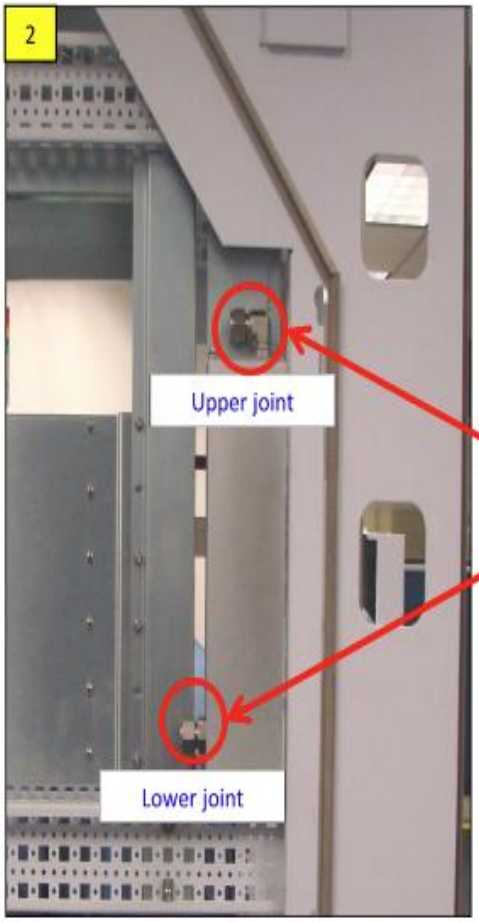
METHOD: **HFE Steel Tubing Preparation Before Assembly**

1. Showing manifold installed. View from the outside of the cabinet.

NOTE: DO NOT TIGHTEN THE COMPRESSION NUTS AT THESE TIME.

2. As shown, loosen the 2 pressure joint nuts on the upper and lower parts of the Manifold block gently by using 1-inch open end wrench.

3. For upper joint : Loose Joint nut with 1 inch open end wrench at the picture no 3 as shown above.



Loose nut

For upper joint : Just Loosen Joint nut only

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	639-582-01	STEEL FLEXIBLE TUBING (LOWER JOINT)	1
2	639-583-01	STEEL FLEXIBLE TUBING (UPPER JOINT)	1

MACHINE: 1 Inch Open End Wrench, Torque 1.6Nm, Bit Hex 5mm

METHOD: **HFE Steel Tubing Installation**

4. Take steel tube (639-582-01) & (639-583-01) then assemble Presure Joint Nut that taken from PSU Manifold. Diagram above shown the orientation of Presure Joint Nut. Make sure the orientation of Presure Joint Nut is correctly install as shown in picture above.

Make sure the end of the pipe is inserted into the rail cabinet.

5. Insert both steel pipe (639-582-01) & (639-583-01) into the HFE Manifold. (639-582-01) for lower joint and (639-583-01) for upper joint.

Note : Manually tighten the joint nut

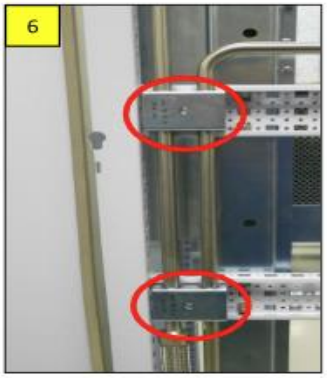
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

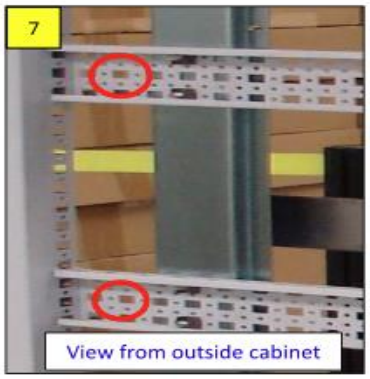
ITEM	P/N	DESCRIPTION	QTY
1	634-175-01	DOUBLE TUBING CLAMP	2
2	602-621-01	SCREW, CAP BUT HD M8X1.25PX45MML SS	2
3	470-501-06	WASHER, FLAT,8.4MM ID, 16 MM OD, SS	2

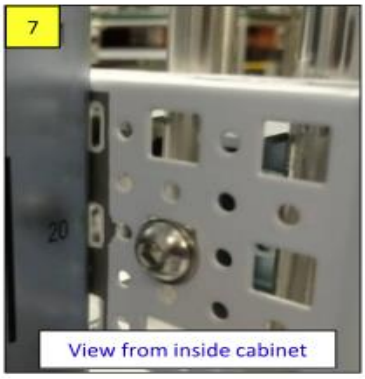
MACHINE: N/A

METHOD: **Double Tubing Clamp Installation**



6. Install 2pcs Double Tubing Clamp (634-175-01) as shown in diagram





7. Diagram shown location of 2pcs Screw (602-621-01) & 2pcs flat washer (470-501-06) Installed from inside cabinet
Note : Manually tighten the screw for next process purpose

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

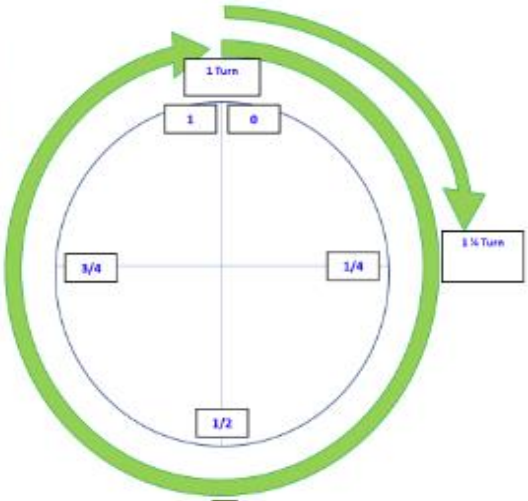
MACHINE: 1 Inch Open End Wrench

METHOD: **Tighten Presure Joint Nut (Upper Joint)**

8. After hand tighten the Presure Joint Nut, **Marked** Upper Presure Joint Nut as shown in diagram.

Next

9. Take 1 Inch Open End Wrench to tighten the Presure Joint Nut **1 ¼ turn**



Next

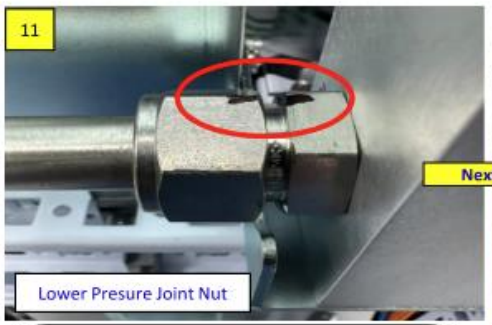
10. Diagram shown After tighten Presure Joint Nut untill **1** ¼

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 1 Inch Open End Wrench

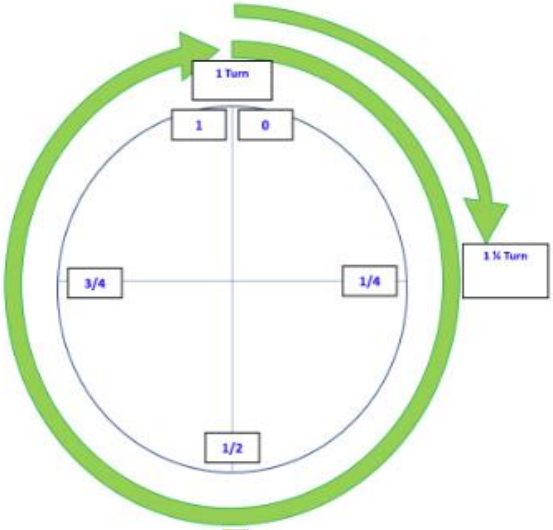
METHOD: **Tighten Presure Joint Nut (Lower Joint)**



11. After hand tighten the Presure Joint Nut, **Marked** Lower Presure Joint Nut as shown in diagram.

Next

12. Take 1 Inch Open End Wrench to tighten the Presure Joint Nut **1 ¼ turn**



Next

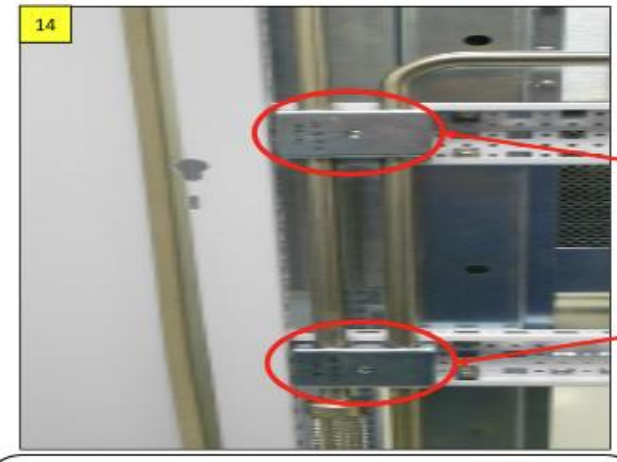
13. Diagram shown After tighten Presure Joint Nut untill **1 ¼**

MAN: CERTIFIED OPERATOR, TECHNICIAN

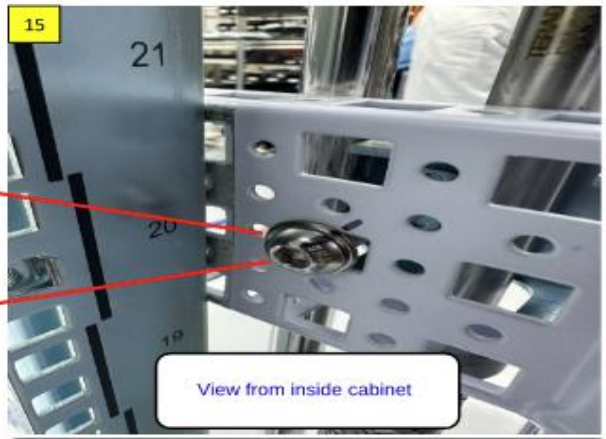
MATERIAL: N/A

MACHINE: Torque 1.6Nm, Bit Hex 5mm, Open End Torque 6.8Nm, 1 Inch open end

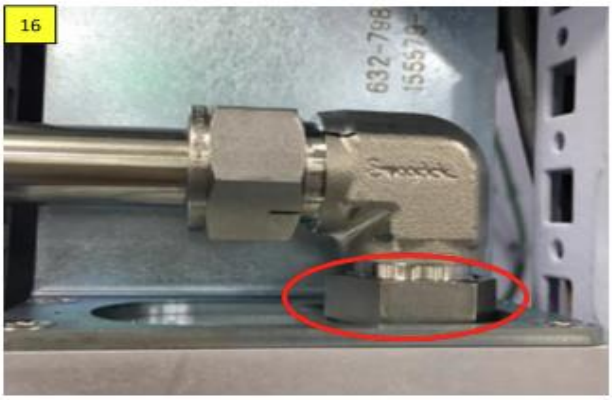
METHOD:



14. Fix 2pcs bracket on the guide rail as shown in figure. Use 5mm Hex Bitt with Torque value of 1.6Nm



15. Add **Torque Marking** with Sharpie Marker after done torque 1.6Nm. Marked **2 places** as picture above.



16. Tighten the upper compression nut with 1 inch Open End Adapter and Open End Torque 6.8Nm

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD: **Final Inspection of Installing HFE Steel Tube**

1. Make sure all nut and screw were fully tighten and marked according to previous assembly
2. Ensure that the HFE Steel Tube and the guide rail are level as shown in the figure.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 1 Inch Open End Wrench

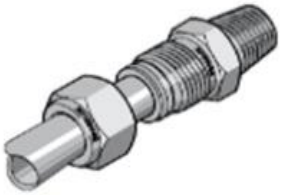
METHOD: **Tighten Presure Joint Nut**

1. Reassemble the joint sleeve. (For rework activity)
2. You may disassemble or reassemble the joint casing several times. Please use this page to install.

Note: Before removing the joint casing part, it must be ensured that the system has been decompressed.

Before disassembly, make a mark and draw a line on the nut and sleeve joint at the back of the pipe.

The purpose of making these marks is to ensure that the nuts can be returned to their original positions when they are reassembled and screwed back.



Insert the pipe with the pre-installed metal ferrule into the sleeve joint until the front metal ferrule is against the joint.

While using a wrench to clamp the casing joint, use another wrench to rotate the nut to the previously tightened position, keeping it flush with the mark made before. At the same time, you will feel the resistance increase significantly, tighten the nut gently.

MAN: CERTIFIED OPERATOR, TECHNICIAN

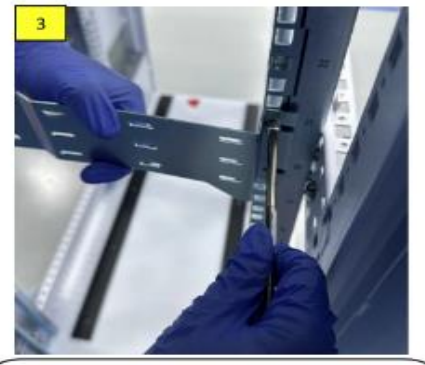
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	631-823-00	CDU "Z" Bracket	1
2	472-426-12	M5 Screws	4

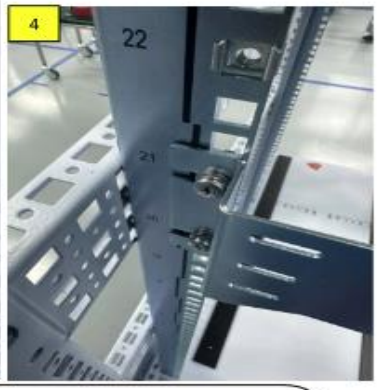
METHOD: **CDU "Z" Bracket Installation**

1. At **Maintanance Side**. Install 2pcs M5 Screws (472-426-12) at location **21M** and **20M** one side only.

2. Insert 1pcs CDU Z Bracket (631-823-00) to the screw that installed before.



3. Insert another 2pcs M5 Screw (472-426-12) at other side of cabinet rail location **21M** and **20M**. Do Not Tighten 4pcs Screw Yet



4. Diagram above show installed CDU Z Bracket (631-823-00) and the orientation.
Note: Leave the Screw Loose. They will be torque later

MAN: CERTIFIED OPERATOR, TECHNICIAN

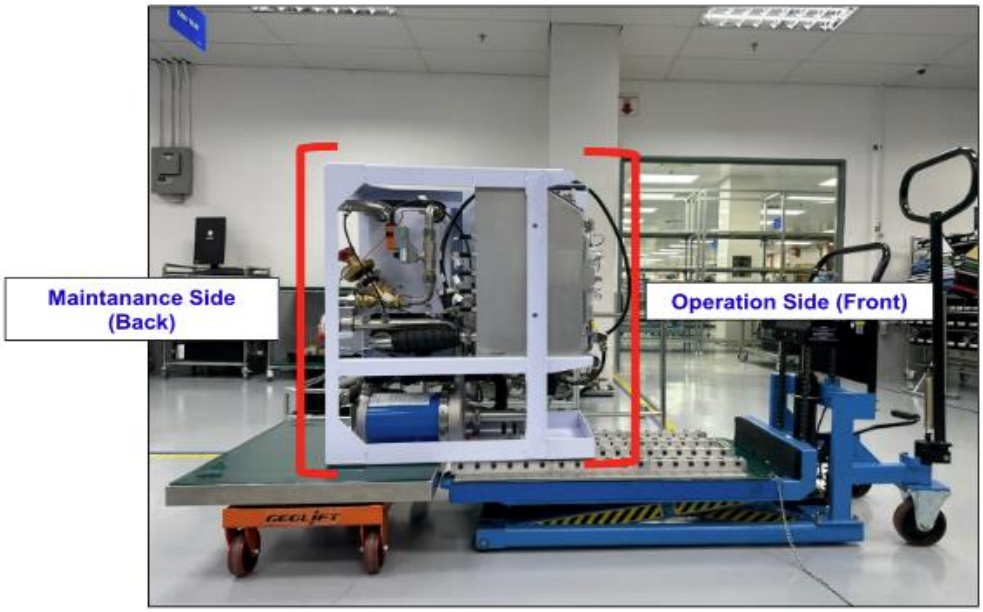
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	660-825-00	CDU ASSEMBLY	1

MACHINE: CDU LIFTER

METHOD: **Preparation CDU Sub Assembly to Lifter**

1. Prepare the Support Cabinet facing Operation Side before installing CDU Assembly as shown in figure.
2. Identify side of the CDU Assembly Operation Side (Front) and Maintanance Side (Back)
3. Slide the CDU to lifter from Operation Side (Front) first, facing towards lifter handle and make sure the power cable not hanging outside of the CDU chasis.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD: **Installation CDU Sub Assembly to Cabinet**

1. While the Support Cabinet facing **Operation Side**. Put Lifter as shown in the picture.
2. Adjust the lifter parallel with Support Cabinet floor.
3. Push the CDU Assembly carefully into the cabinet untill CDU chasis touch the Cabinet CDU Slider.

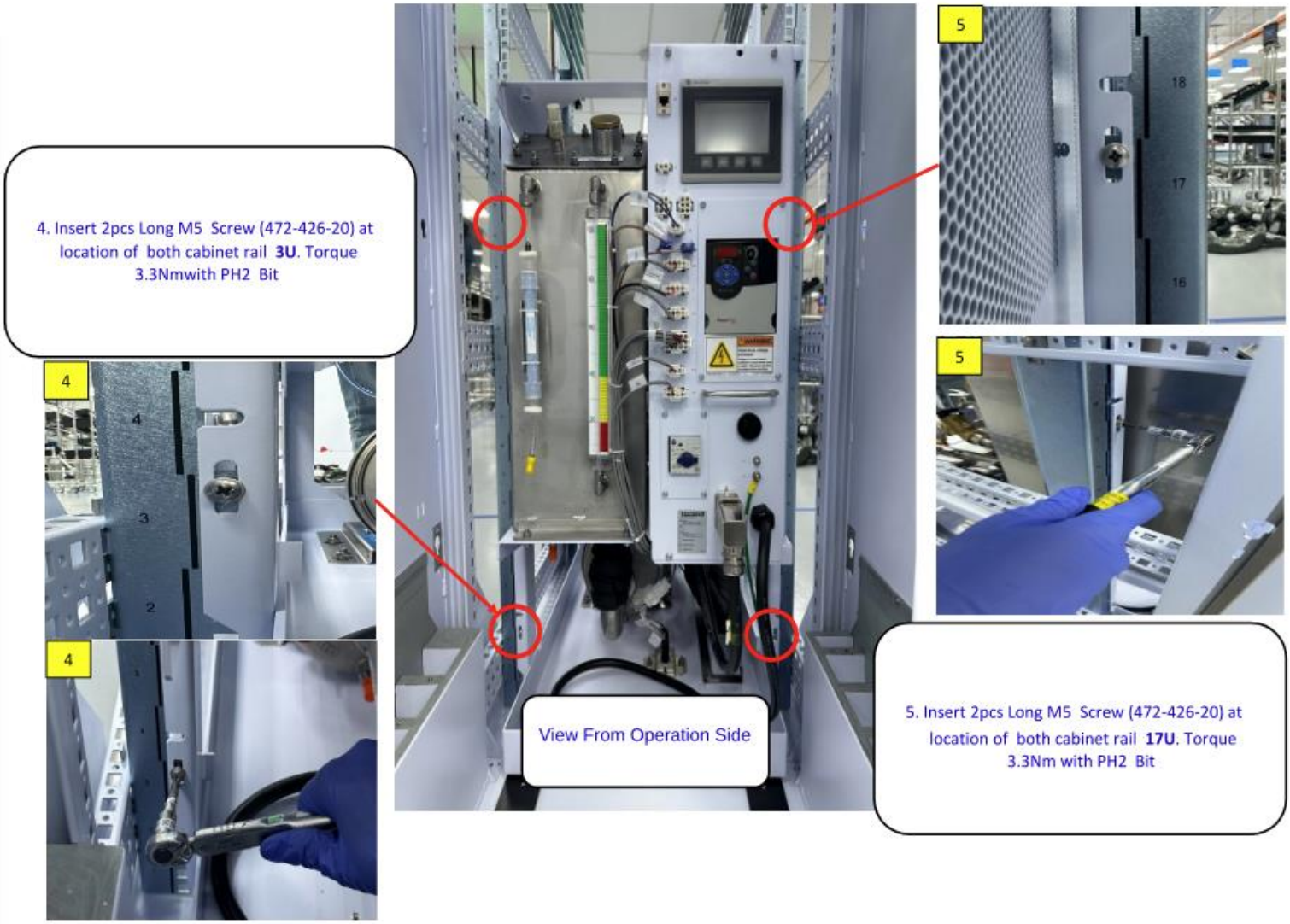
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	472-426-20	M5 Long Screws	4

MACHINE: Torque 3.3Nm, PH2 Bit

METHOD: **Installation CDU Sub Assembly to cabinet.**



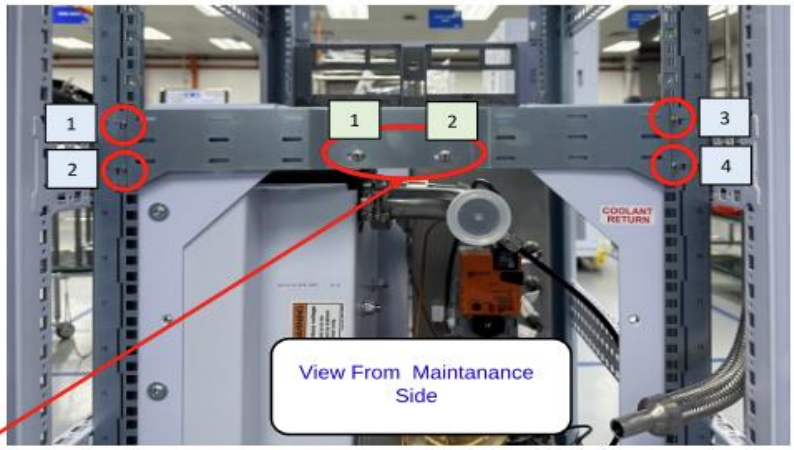
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
2	472-426-12	M5 Screws	2

MACHINE: Torque 3.3Nm, PH2 Bit

METHOD: **Installation CDU Sub Assembly to cabinet.**



6. Insert 2pcs M5 Screw (472-426-12) at center of CDU Z Bracket that installed before and Torque to 3.3Nm with PH2 Bit

7. Torque 4pcs (472-426-12) that installed before with 3.3Nm and PH2 Bit and **marked the screw** as diagram above.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

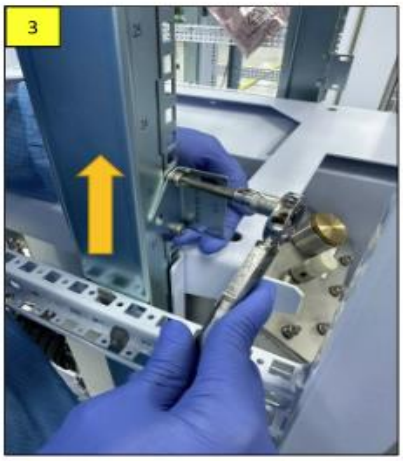
ITEM	P/N	DESCRIPTION	QTY
1	629-538-00	ADB Shelf Bracket	2
2	472-426-12	M5 Screws	4

MACHINE: Torque 3.3Nm, PH2 Bit

METHOD: **Installation ADB Shelf Bracket**

1. From Operation Side, install ADB Shelf Bracket (629-538-00) to the cabinet rail **22M** and **23M** with 2 pcs M5 Screws (472-426-12) each bracket.

2. Picture above shown the orientation of ADB Shelf Bracket (629-538-00).



3. **Push up the bracket** while torque the ADB Shelf Bracket (629-538-00) - torque to 3.3Nm.

MAN: CERTIFIED OPERATOR, TECHNICIAN

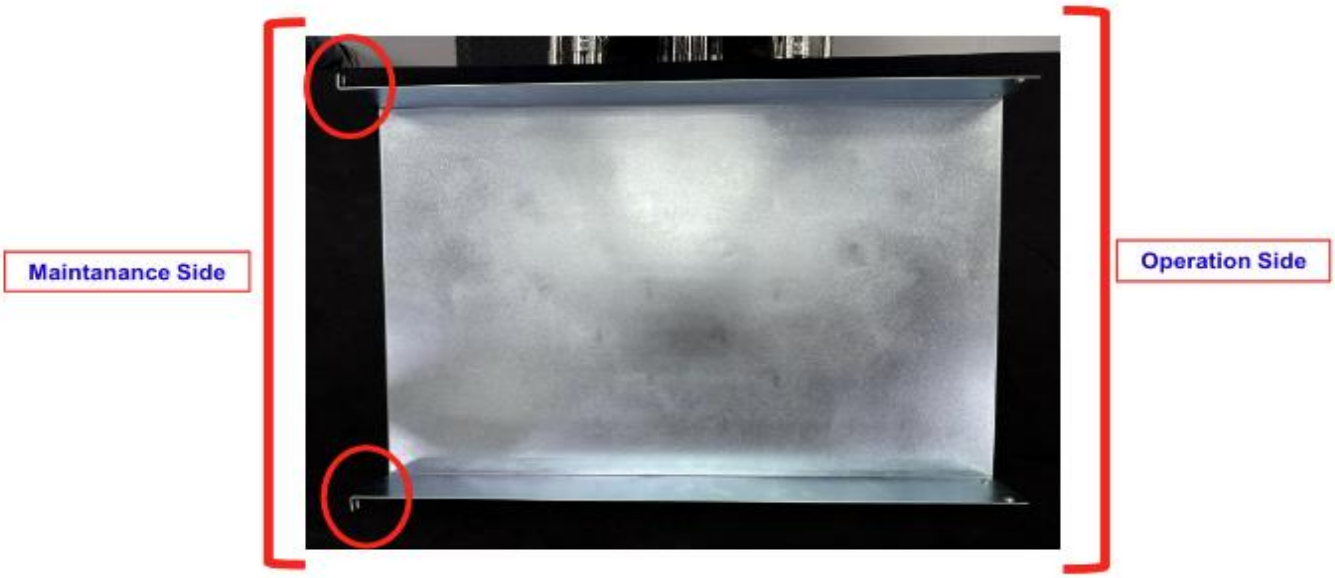
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	629-537-00	ADB Shelf	1

MACHINE: Torque 3.3Nm, PH2 Bit

METHOD: **ADB Shelf Preparation**

1. Take ADB Shelf (629-537-00) to work table.
2. As shown in diagram, verify the position to hold the ADB Shelf before installing to cabinet.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	472-426-12	M5 Screw	8

MACHINE: Torque 3.3Nm, PH2 Bit

METHOD: **Installation ADB Shelf**

3. From **Maintanance Side** of cabinet, Insert the ADB Shelf carefully untill ADB shelf touching the cabinet rail

4. Install 4 pcs M5 Screw (472-426-12) on both side of cabinet rail at location 22M and 23M.
5. Lift up the ADB Shelf while torque the 4 pcs M5 Screws - Torque to 3.3Nm and PH2 Bit

6. Install 4 pcs M5 Screw (472-426-12) on both side of ADB Shelf Bracket at location 22M and 23M.
7. Lift up the ADB Shelf while torque the 4 pcs M5 Screws - Torque to 3.3Nm and PH2 Bit